

Proposal to change MIPD Topic

Previous topic

MIPD 2026 Project Endorsement Form

STUDENT INFORMATION			
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Specialisation	OFFSHORE ENGINEERING	Student ID	2402280
IWSP Company	Seatrium		

PROJECT INFORMATION	
Poster Title	Mobile Offshore Solar Unit (MOSU): A Vessel Concept for Autonomous Floating Solar Farm Deployment
Motivation	Interning at Seatrium's Offshore Energies Engineering Department, provide insight into current offshore energy projects, leading to the conceptualization of a mobile and self-deploying solar vessel designed to increase energy harvesting capability through the rapid deployment of floating solar farms.
Objectives, Scope of Work, and Methodolgy	Objectives: - Conceptualize a MOSU through CAD softwares or sketches - Research on advantages that the vessel and how it can improve solar energy havesting capabilities. - Provide suggestions on future studies that can be done on this concept idea (ie; stability studies, maintenance studies) Scope of Work: - Project is related to design engineering for the offshore environment which is related to my internship department, company and my specialization. Methodology: - 3D CAD Design, Sketches, Research on Solar Energy Efficiency
	Layers of solar panels are stacked from the bottom hull to several metres

Mobile Offshore Solar Unit: A Vessel Concept for Autonomous Floating Solar Farm

New Proposed Topic



Powering the Ocean's Future: A Topology-Driven Hydrodynamic Study for Floating Nuclear Power Platforms

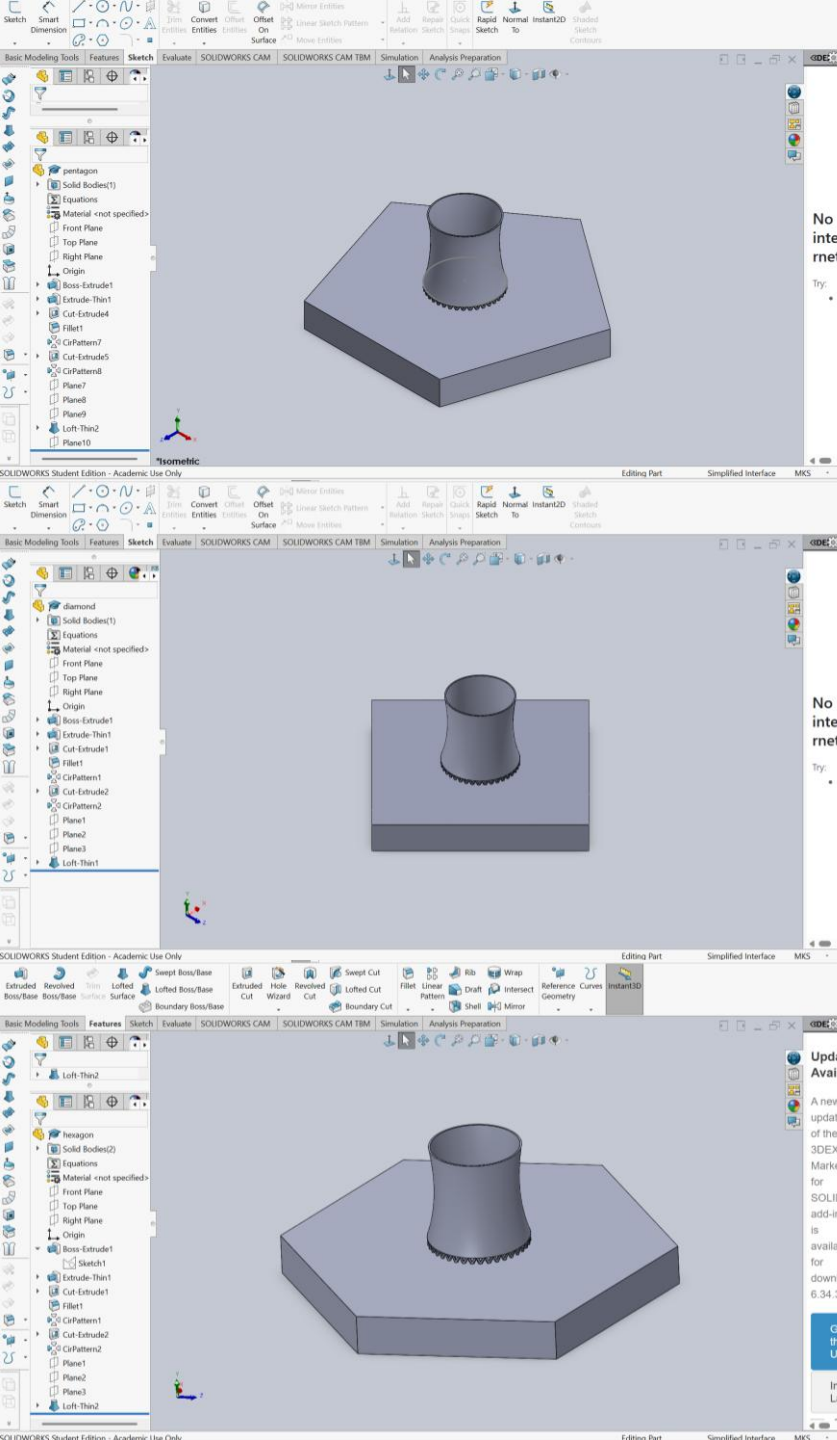
Topologies

To study to response of square, diamond, pentagon and hexagon shapes floating nuclear facilities ocean waves.

Primary study: RAO vs frequency response

Secondary studies: RAO vs time response, Mooring studies, Effect of air flow into nuclear tower (circular cylinder fluid mechanics concept), nuclear studies

Considerations: Easy of scale (hexagon = honecomb structure,, but at what cost), environmental (replace the “Western Islands”), financial and economic



Strong case

- Novel idea
- Quantitative concepts closely linked to fluid mechanics, naval architecture and offshore engineering
- Recently government announced plans to make the “Western Islands”,, this is a new concept
- Between my first proposal and this, I have picked up Solidworks, NAPA, Ansys AQWA, a little bit of Ansys Fluent.
- High potential for future studies. Impactful.
- Interning at Seatrium

Powering the Ocean's Future: A Topology-Driven Hydrodynamic Study for Floating Nuclear Power Platforms

Please consider my MIPD project idea change.

Regards
Danish (2402280)